EXP 8: Write a program to implement Linear Regression (LR) algorithm in python

CODING:

```
# Linear Regression Algorithm

# Training Dataset
X = [1, 2, 3, 4, 5]
Y = [10, 20, 30, 40, 50]

# Calculate Mean
n = len(X)

mean_x = sum(X) / n
mean_y = sum(Y) / n

# Calculate Slope
numerator = 0
denominator = 0

for i in range(n):
    numerator += (X[i] - mean_x) * (Y[i] - mean_y)
    denominator += (X[i] - mean_x) ** 2

slope = numerator / denominator

# Calculate Intercept
intercept = mean_y - slope * mean_x
```

```
# Prediction
predictions = []

for x in X:
    y_pred = slope * x + intercept
    predictions.append(y_pred)

# Display Results
print("Training Data:")
print("X Values:", X)
print("Y Values:", Y)

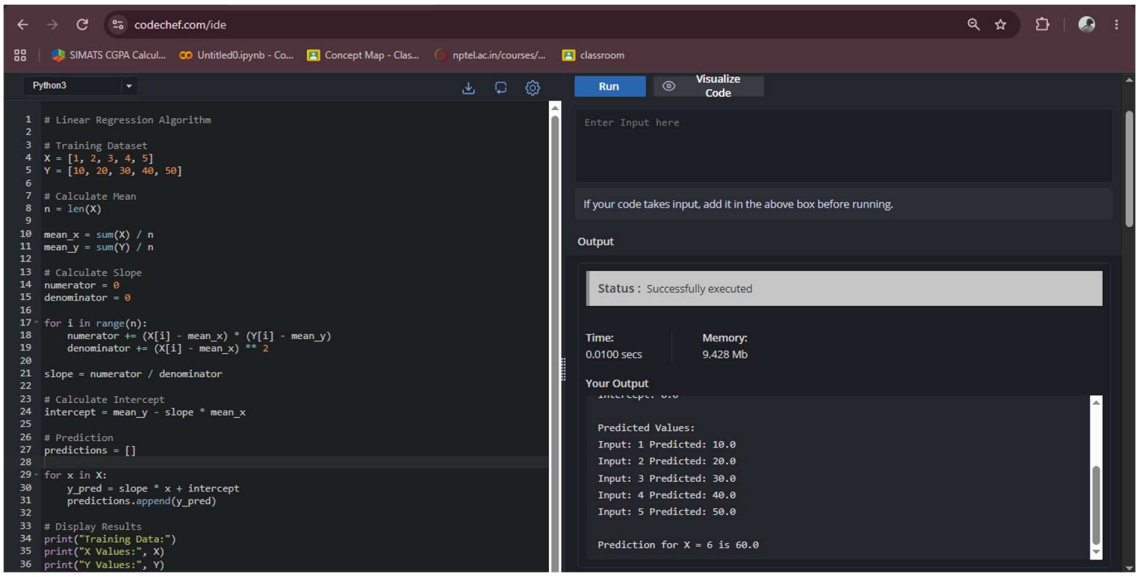
print("\nLinear Regression Model")
print("Slope:", slope)
print("Intercept:", intercept)

print("\nPredicted Values:")
for i in range(n):
    print("Input:", X[i], "Predicted:", predictions[i])

# Predict New Value
new_x = 6
new_y = slope * new_x + intercept

print("\nPrediction for X =", new_x, "is", new_y)
```

OUTPUT:




```python
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